

COMPUTER MAINTENANCE

FULL VISUAL STUDY MANUAL

Computer Maintenance Cycle

Prevent problems before they interrupt work

Inspect

Check condition and symptoms

Clean

Keep safe external areas clean

Update

Apply approved software updates

Back up

Protect important data

Test

Confirm normal operation

Record

Document maintenance performed

Course Code	EDA-MNT-110
Level	Beginner to Intermediate
Duration	6 Weeks (19 Lessons)
Certificate	Professional Certificate in Computer Maintenance
Prerequisite	Computer Hardware Fundamentals
Learning Format	Classroom + Safe Guided Practicals + Maintenance Planning Project

Safety standard: this beginner course focuses on preventive care, safe external checks and approved software maintenance. Electrical repair, damaged batteries, exposed components and hazardous internal servicing must be handled by qualified personnel.

Course Introduction

Computer Maintenance teaches learners how to keep computers clean, organised, updated, backed up, secure and reliable. It covers preventive maintenance, safe cleaning, software updates, storage management, backups, performance monitoring, browser care, battery and peripheral care, troubleshooting, maintenance records and planning.

Learning Objectives

- Explain preventive and corrective computer maintenance.
- Prepare a safe maintenance workspace.
- Perform safe external cleaning and inspection.
- Use approved operating-system and application maintenance tools.
- Manage storage and backups responsibly.
- Monitor basic performance without unsafe system changes.
- Maintain simple asset and maintenance records.
- Apply a structured troubleshooting and escalation workflow.
- Create a preventive-maintenance plan for a computer lab.

Course Structure

1. Lesson 1: Introduction to Computer Maintenance
2. Lesson 2: Maintenance Safety and Preparation
3. Lesson 3: External Cleaning of Computers
4. Lesson 4: Keyboard, Mouse and Peripheral Care
5. Lesson 5: Operating System Updates
6. Lesson 6: Application Updates and Software Hygiene
7. Lesson 7: Storage Management and Disk Space
8. Lesson 8: Backup Fundamentals
9. Lesson 9: File Organisation and Version Control
10. Lesson 10: Performance Monitoring Basics
11. Lesson 11: Startup and Shutdown Good Practice
12. Lesson 12: Basic Malware Prevention and Security Maintenance
13. Lesson 13: Browser Maintenance
14. Lesson 14: Laptop Battery and Power Care
15. Lesson 15: Printer and Scanner Maintenance Basics
16. Lesson 16: Maintenance Logs and Asset Records
17. Lesson 17: Safe Troubleshooting Workflow
18. Lesson 18: Maintenance Schedule Planning
19. Lesson 19: Final Computer Maintenance Project

Lesson 1: Introduction to Computer Maintenance

Introduction

Computer maintenance is the planned care of hardware, software and data so a computer remains reliable, secure and useful. Preventive maintenance aims to reduce avoidable problems before they interrupt work.

Learning Objectives

- Explain the key principles of introduction to computer maintenance.
- Complete the safe guided practical for introduction to computer maintenance.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Maintenance includes inspection, safe cleaning, updates, storage management, backups, performance checks and record keeping.

Key point: Preventive maintenance is scheduled before a failure; corrective maintenance responds after a problem appears.

Key point: Beginner learners should focus on safe external care and approved software tools.

Key point: Internal electrical repair, damaged batteries and exposed power components require qualified personnel.

Computer Maintenance Cycle

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Check condition and symptoms

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Document maintenance performed

Figure 1: Instructional visual for Introduction to Computer Maintenance.

Step-by-Step Guided Practical

1. Inspect an authorised computer without opening it.
2. Record visible condition, free storage and update status.
3. List three preventive tasks and three corrective tasks.
4. Create a simple maintenance checklist.

Ethan Digital Academy Case Study

Ethan Digital Academy can schedule routine checks for lab computers so training is not repeatedly interrupted by avoidable issues.

Review Questions

20. What is the purpose of introduction to computer maintenance?
21. State two good maintenance practices from this lesson.
22. What unsafe or unnecessary action should be avoided?
23. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 1 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 2: Maintenance Safety and Preparation

Introduction

Safe maintenance begins before any cleaning or troubleshooting. A dry workspace, correct shutdown and clear limits reduce risk to both people and equipment.

Learning Objectives

- Explain the key principles of maintenance safety and preparation.
- Complete the safe guided practical for maintenance safety and preparation.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Save work and use normal operating-system shutdown.

Key point: Disconnect external power only when the instructor or manufacturer procedure requires it.

Key point: Never work on wet equipment or damaged electrical cables.

Key point: Do not open power supplies or other hazardous electrical components.

Key point: Use manufacturer guidance for model-specific service procedures.

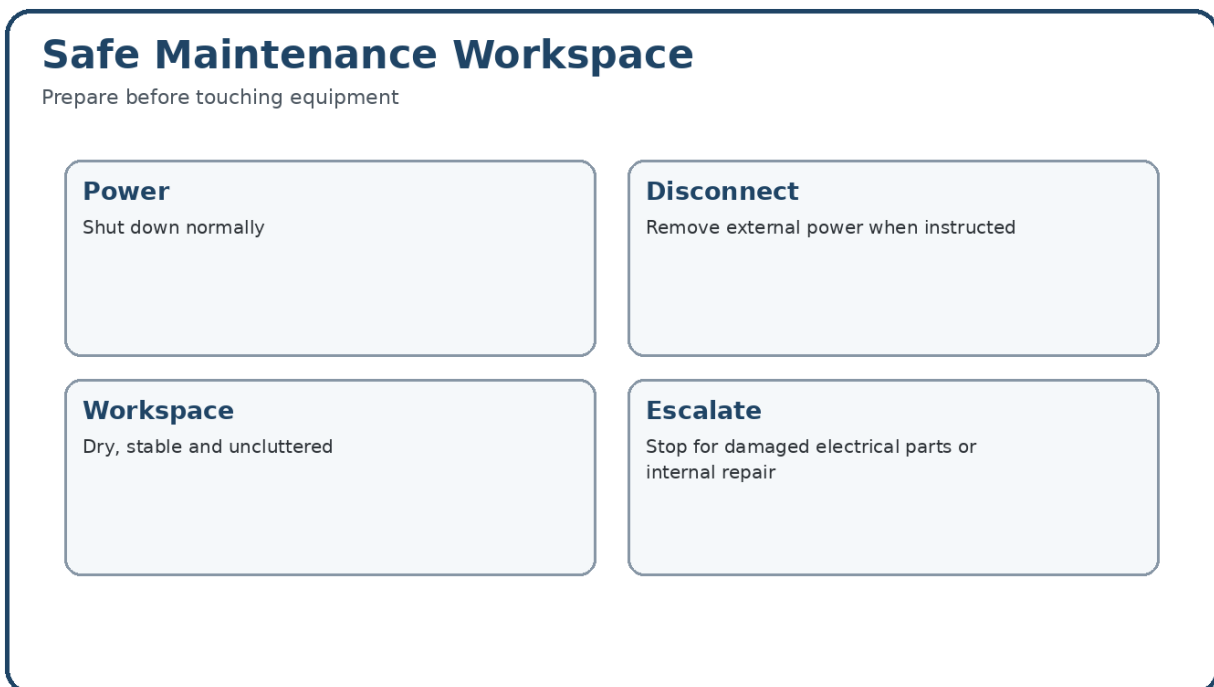


Figure 2: Instructional visual for Maintenance Safety and Preparation.

Step-by-Step Guided Practical

1. Save and close all applications.
2. Shut down the computer normally.
3. Inspect the external power cable without touching damaged areas.
4. Prepare a dry, uncluttered workspace.
5. Write when a learner should stop and call a technician.

Ethan Digital Academy Case Study

Academy learners should know the difference between a safe external check and a repair that belongs to trained technical staff.

Review Questions

24. What is the purpose of maintenance safety and preparation?
25. State two good maintenance practices from this lesson.
26. What unsafe or unnecessary action should be avoided?
27. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 2 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 3: External Cleaning of Computers

Introduction

Dust, fingerprints and debris can affect comfort, airflow and appearance. Cleaning must be done with the equipment in a safe state and with suitable materials.

Learning Objectives

- Explain the key principles of external cleaning of computers.
- Complete the safe guided practical for external cleaning of computers.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use a soft lint-free cloth for external surfaces.

Key point: Do not spray liquid directly into ports, keyboards or vents.

Key point: Screens require gentle cleaning with a method suitable for the display manufacturer.

Key point: Loose keyboard debris can be removed using safe, non-damaging methods.

Key point: Internal cleaning should be performed only on prepared training equipment by authorised personnel.

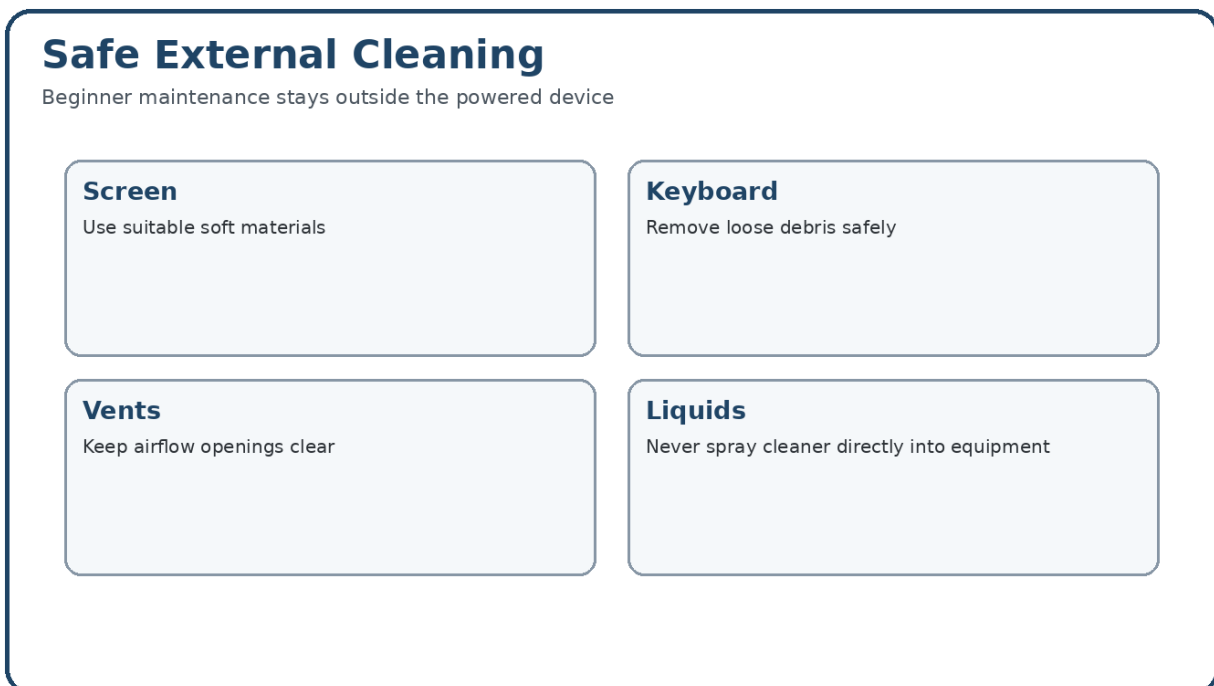


Figure 3: Instructional visual for External Cleaning of Computers.

Step-by-Step Guided Practical

1. Power down the device.
2. Wipe the external case with a dry or appropriately prepared soft cloth.
3. Clean the screen gently according to instructor guidance.
4. Inspect vents and keyboard for loose debris.
5. Confirm no moisture remains before normal use.

Ethan Digital Academy Case Study

A clean Academy lab supports professional presentation and helps keep ventilation openings unobstructed.

Review Questions

28. What is the purpose of external cleaning of computers?
29. State two good maintenance practices from this lesson.
30. What unsafe or unnecessary action should be avoided?
31. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 3 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 4: Keyboard, Mouse and Peripheral Care

Introduction

Peripherals are handled frequently and benefit from routine inspection and cleaning.

Learning Objectives

- Explain the key principles of keyboard, mouse and peripheral care.
- Complete the safe guided practical for keyboard, mouse and peripheral care.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Disconnect removable peripherals when appropriate before cleaning.

Key point: Check cables for visible wear, bent connectors or loose connections.

Key point: Clean mouse surfaces and keyboard exteriors without flooding openings.

Key point: Printers and scanners should be maintained according to manufacturer instructions.

Key point: Damaged cables or power adapters should be removed from use and reported.

Step-by-Step Guided Practical

1. Inspect a keyboard and mouse.
2. Clean external surfaces safely.
3. Check USB cable and connector condition.
4. Reconnect the device and confirm normal operation.
5. Record the result in a maintenance log.

Ethan Digital Academy Case Study

Academy computer-lab peripherals can be checked weekly because frequent learner use increases wear.

Review Questions

32. What is the purpose of keyboard, mouse and peripheral care?
33. State two good maintenance practices from this lesson.
34. What unsafe or unnecessary action should be avoided?
35. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 4 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 5: Operating System Updates

Introduction

Operating-system updates can correct bugs, improve compatibility and address security issues. Maintenance should use official update mechanisms.

Learning Objectives

- Explain the key principles of operating system updates.
- Complete the safe guided practical for operating system updates.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use the operating system's normal update settings.

Key point: Review whether an update requires a restart.

Key point: Save work before restarting.

Key point: Do not download unofficial 'update tools' from random websites.

Key point: Organisations may schedule updates to avoid interrupting classes.

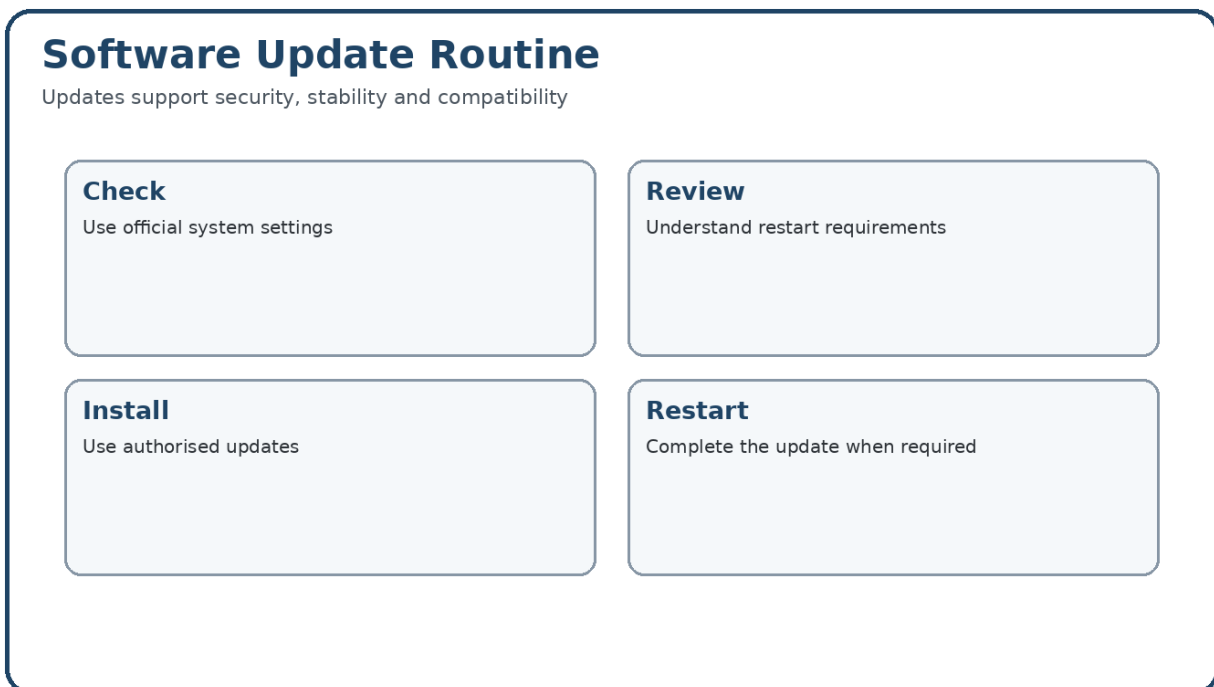


Figure 5: Instructional visual for Operating System Updates.

Step-by-Step Guided Practical

1. Open the official update settings.
2. Check update status without bypassing administrator controls.
3. Record whether a restart is pending.
4. If authorised, install normal updates.
5. Confirm the computer starts and key applications open normally.

Ethan Digital Academy Case Study

The Academy can schedule updates outside teaching hours and verify classroom software afterward.

Review Questions

36. What is the purpose of operating system updates?
37. State two good maintenance practices from this lesson.
38. What unsafe or unnecessary action should be avoided?
39. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 5 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 6: Application Updates and Software Hygiene

Introduction

Applications also require maintenance. Unsupported or unnecessary software can create compatibility, storage and security problems.

Learning Objectives

- Explain the key principles of application updates and software hygiene.
- Complete the safe guided practical for application updates and software hygiene.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Update applications through trusted vendor or app-store channels.

Key point: Remove software only when authorised and no longer required.

Key point: Avoid installing unknown 'cleaner' or 'booster' programs promising unrealistic speed improvements.

Key point: Keep licences and account access organised.

Key point: Check that important teaching applications still work after updates.

Step-by-Step Guided Practical

1. List five installed learning applications.
2. Identify their normal update source.
3. Check one application's version using its About screen.
4. Record unused software for administrator review rather than deleting it without permission.
5. Test one key application after an authorised update.

Ethan Digital Academy Case Study

Academy administrators can maintain a standard list of approved applications for each training room.

Review Questions

40. What is the purpose of application updates and software hygiene?
41. State two good maintenance practices from this lesson.
42. What unsafe or unnecessary action should be avoided?

43. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 6 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 7: Storage Management and Disk Space

Introduction

Computers need adequate free storage for updates, temporary files and user work. Good housekeeping reduces clutter without deleting important data.

Learning Objectives

- Explain the key principles of storage management and disk space.
- Complete the safe guided practical for storage management and disk space.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use built-in storage tools to see which categories consume space.

Key point: Organise files into clear folders.

Key point: Delete only files you own and understand.

Key point: Emptying a recycle bin can make deletion harder to reverse, so review first.

Key point: Back up important work before major cleanup.

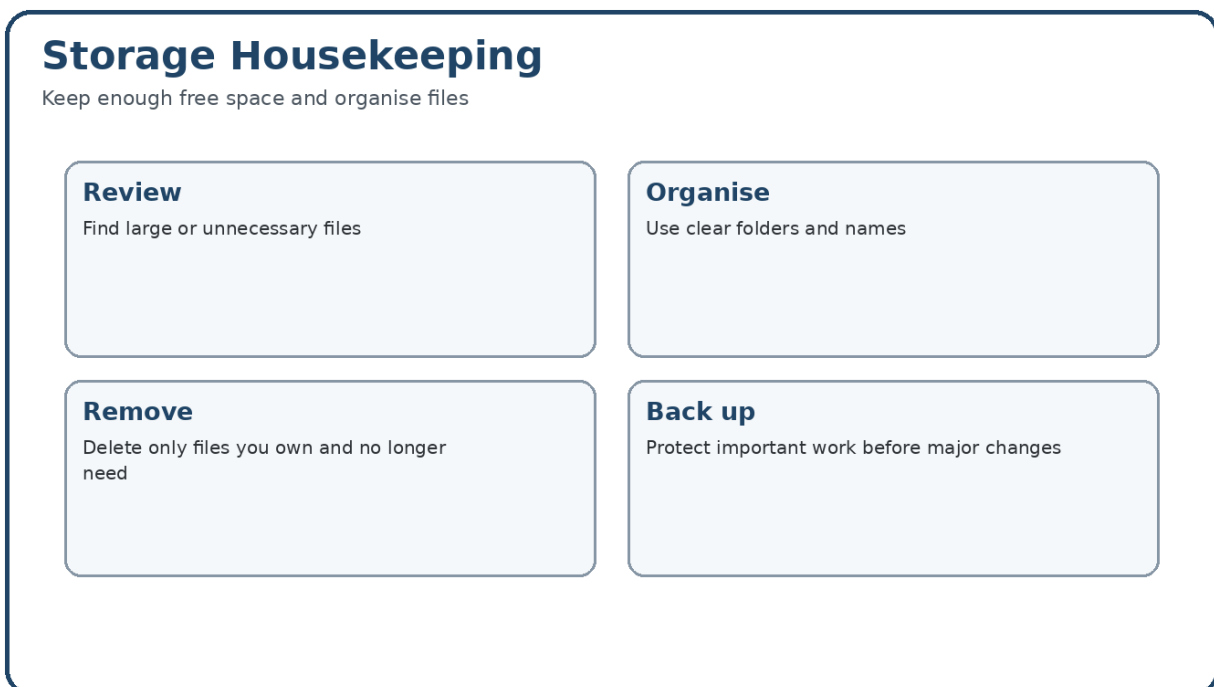


Figure 7: Instructional visual for Storage Management and Disk Space.

Step-by-Step Guided Practical

1. Open the system storage view.
2. Record total and free storage.
3. Identify one folder containing harmless practice files.
4. Remove only an instructor-approved unnecessary practice file.
5. Record free space again.

Ethan Digital Academy Case Study

Academy lab machines can use standard folders and periodic review so old downloads do not consume all available storage.

Review Questions

44. What is the purpose of storage management and disk space?
45. State two good maintenance practices from this lesson.
46. What unsafe or unnecessary action should be avoided?
47. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 7 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 8: Backup Fundamentals

Introduction

A backup is a separate copy of important data used for recovery after deletion, hardware failure or another problem.

Learning Objectives

- Explain the key principles of backup fundamentals.
- Complete the safe guided practical for backup fundamentals.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: A backup should be separate from the working copy.

Key point: The 3-2-1 idea encourages multiple copies, more than one storage type and one separate copy.

Key point: Cloud sync and backup are related but not always identical; accidental changes may synchronise too.

Key point: Backups should be protected from unauthorised access.

Key point: A restore test confirms that the backup can actually be used.

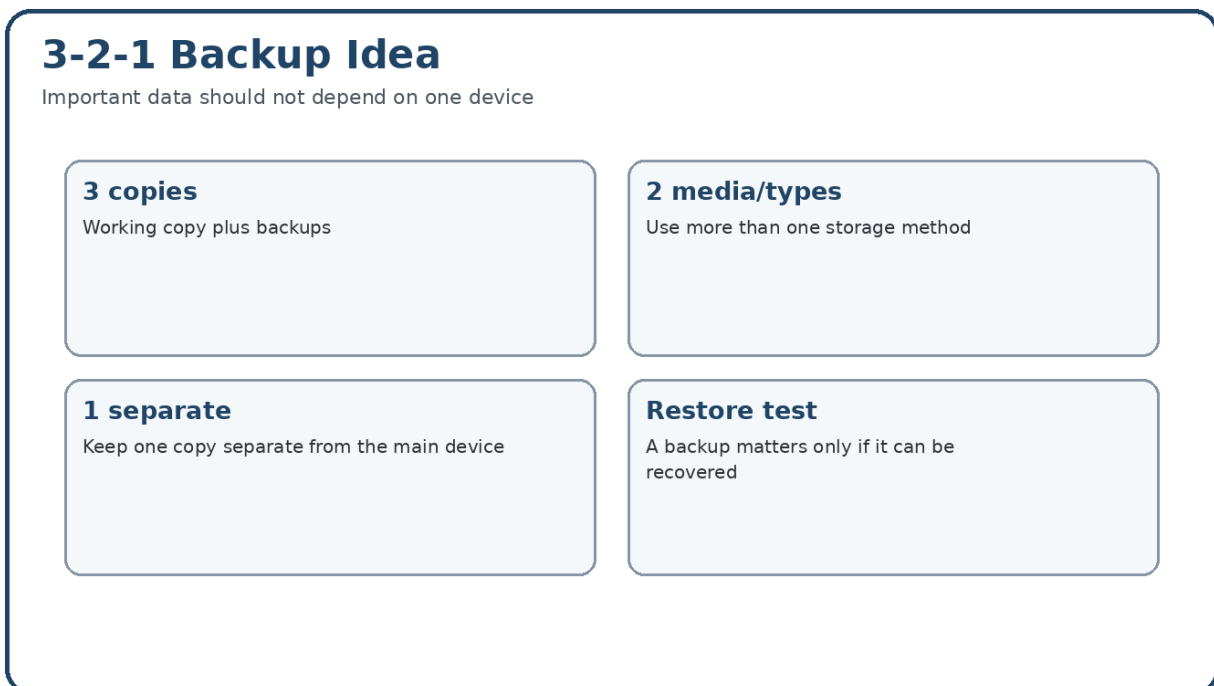


Figure 8: Instructional visual for Backup Fundamentals.

Step-by-Step Guided Practical

1. Create a harmless practice folder.
2. Copy it to an authorised backup location.
3. Rename one file in the working copy.
4. Restore the original from the backup into a separate test folder.
5. Compare both copies.

Ethan Digital Academy Case Study

Ethan Digital Academy should protect course materials and administrative records with a documented backup routine.

Review Questions

48. What is the purpose of backup fundamentals?
49. State two good maintenance practices from this lesson.
50. What unsafe or unnecessary action should be avoided?
51. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 8 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 9: File Organisation and Version Control

Introduction

Maintenance also means keeping information organised. Poor naming and uncontrolled duplicates make recovery and collaboration difficult.

Learning Objectives

- Explain the key principles of file organisation and version control.
- Complete the safe guided practical for file organisation and version control.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use meaningful file names with subject and date or version where useful.

Key point: Keep an editable master separate from exported copies.

Key point: Avoid confusing names such as final-final-new.

Key point: Archive completed work according to organisational rules.

Key point: Do not store passwords inside ordinary documents.

Step-by-Step Guided Practical

1. Create folders named Current, Archive and Exports.
2. Rename three fictional files clearly.
3. Move an old practice copy to Archive.
4. Export one document to PDF and place it in Exports.
5. Explain which file is the editable master.

Ethan Digital Academy Case Study

Academy course teams can keep approved masters separate from learner-facing PDF exports.

Review Questions

52. What is the purpose of file organisation and version control?
53. State two good maintenance practices from this lesson.
54. What unsafe or unnecessary action should be avoided?
55. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 9 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 10: Performance Monitoring Basics

Introduction

Slow performance can come from many causes. Maintenance starts by identifying the symptom and checking normal resource information rather than making random changes.

Learning Objectives

- Explain the key principles of performance monitoring basics.
- Complete the safe guided practical for performance monitoring basics.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Task Manager or similar tools can show CPU, memory, storage and application activity.

Key point: High usage during a demanding task is not automatically a fault.

Key point: Too many applications can reduce available memory.

Key point: Very low free storage can affect updates and normal operation.

Key point: Avoid changing advanced system settings unless authorised and understood.

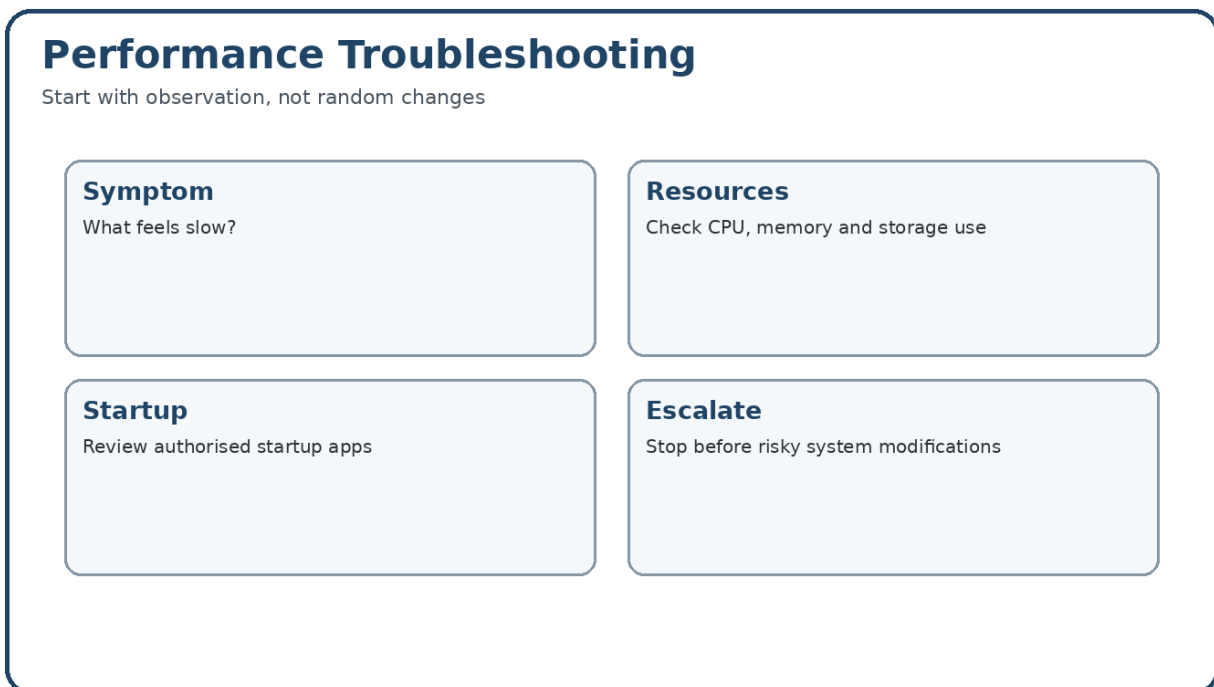


Figure 10: Instructional visual for Performance Monitoring Basics.

Step-by-Step Guided Practical

1. Open Task Manager or the system monitor.
2. Observe CPU and memory at idle.
3. Open a normal application and observe the change.
4. Close it and confirm resource use settles.
5. Record observations without ending unknown system processes.

Ethan Digital Academy Case Study

An instructor can identify whether a slow lab computer is simply running too many applications before requesting technical support.

Review Questions

56. What is the purpose of performance monitoring basics?
57. State two good maintenance practices from this lesson.
58. What unsafe or unnecessary action should be avoided?
59. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 10 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 11: Startup and Shutdown Good Practice

Introduction

Correct startup and shutdown help protect open work and allow the operating system to complete normal processes.

Learning Objectives

- Explain the key principles of startup and shutdown good practice.
- Complete the safe guided practical for startup and shutdown good practice.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use the operating system's shutdown command rather than routinely cutting power.

Key point: Restart is useful after some updates or temporary software issues.

Key point: Sleep preserves a session but is not the same as shutdown.

Key point: Unexpected power loss can interrupt writes and cause data loss.

Key point: A UPS can provide time for safe shutdown during short outages.

Step-by-Step Guided Practical

1. Identify Shut down, Restart and Sleep options.
2. Explain when each is appropriate.
3. Save a practice file.
4. Restart the computer if authorised.
5. Reopen the file and confirm it is intact.

Ethan Digital Academy Case Study

Academy labs should use normal shutdown at the end of sessions rather than switching off wall power while computers are running.

Review Questions

60. What is the purpose of startup and shutdown good practice?
61. State two good maintenance practices from this lesson.
62. What unsafe or unnecessary action should be avoided?

63. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 11 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 12: Basic Malware Prevention and Security Maintenance

Introduction

Security maintenance reduces exposure to malicious software and unsafe downloads. Learners should use approved defensive tools rather than experimenting with suspicious files.

Learning Objectives

- Explain the key principles of basic malware prevention and security maintenance.
- Complete the safe guided practical for basic malware prevention and security maintenance.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Keep supported operating systems and applications updated.

Key point: Use approved antivirus or built-in security protection.

Key point: Download software from trusted official sources.

Key point: Do not disable security controls to make an unknown file run.

Key point: Report suspicious pop-ups, unexpected account prompts or unusual behaviour.



Figure 12: Instructional visual for Basic Malware Prevention and Security Maintenance.

Step-by-Step Guided Practical

1. Open the built-in security status page.
2. Confirm protection status without changing administrator settings.
3. Review the last update date if shown.
4. Write four signs that should be reported.
5. Explain why an unknown 'free cleaner' download may be risky.

Ethan Digital Academy Case Study

Academy computers should use standard security settings and approved software sources across the lab.

Review Questions

64. What is the purpose of basic malware prevention and security maintenance?
65. State two good maintenance practices from this lesson.
66. What unsafe or unnecessary action should be avoided?
67. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 12 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 13: Browser Maintenance

Introduction

Browsers store history, cookies, cache and extensions. Maintenance should improve reliability without casually deleting information users need.

Learning Objectives

- Explain the key principles of browser maintenance.
- Complete the safe guided practical for browser maintenance.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Keep the browser updated.

Key point: Review extensions and remove only those that are authorised and unnecessary.

Key point: Cache can sometimes be cleared when troubleshooting website display problems.

Key point: Cookies may store sign-in sessions, so clearing them can sign users out.

Key point: Bookmarks should be organised and suspicious extensions reported.

Step-by-Step Guided Practical

1. Open browser settings.
2. Check browser version/update status.
3. Review installed extensions without adding unknown ones.
4. Create a bookmark folder for Academy resources.
5. Write what could happen if cookies are cleared.

Ethan Digital Academy Case Study

A standard browser setup helps Academy learners access the LMS consistently.

Review Questions

68. What is the purpose of browser maintenance?
69. State two good maintenance practices from this lesson.
70. What unsafe or unnecessary action should be avoided?
71. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 13 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 14: Laptop Battery and Power Care

Introduction

Laptop batteries are consumable components. Good power and heat habits can support reliable operation.

Learning Objectives

- Explain the key principles of laptop battery and power care.
- Complete the safe guided practical for laptop battery and power care.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Use a compatible manufacturer-approved charger or suitable certified replacement.

Key point: Avoid blocking ventilation while charging or using the laptop.

Key point: Built-in battery-health features should be used where provided.

Key point: A swollen, leaking, unusually hot or physically damaged battery should not be handled casually; stop using the device and report it.

Key point: Battery replacement procedures vary by model and may require qualified service.

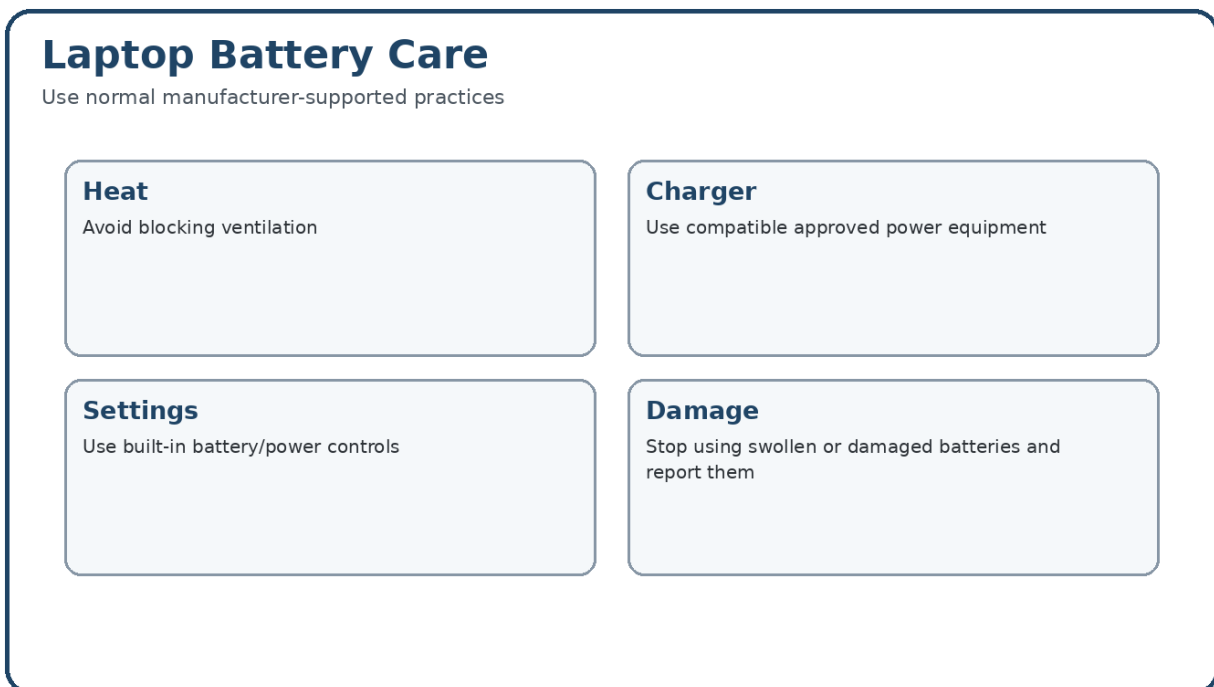


Figure 14: Instructional visual for Laptop Battery and Power Care.

Step-by-Step Guided Practical

1. Open battery/power settings.
2. Record current charge level.
3. Identify available battery-saver or power-mode options.
4. Inspect the laptop exterior for obvious damage without opening it.
5. Write the correct response to a visibly swollen battery: stop use and escalate.

Ethan Digital Academy Case Study

Academy laptops should be stored and charged in suitable conditions and damaged batteries removed from service by authorised staff.

Review Questions

72. What is the purpose of laptop battery and power care?
73. State two good maintenance practices from this lesson.
74. What unsafe or unnecessary action should be avoided?
75. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 14 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 15: Printer and Scanner Maintenance Basics

Introduction

Printers and scanners require routine care, correct consumables and manufacturer-supported troubleshooting.

Learning Objectives

- Explain the key principles of printer and scanner maintenance basics.
- Complete the safe guided practical for printer and scanner maintenance basics.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Keep paper dry and correctly loaded.

Key point: Use suitable ink, toner or other consumables for the model.

Key point: Paper jams should be cleared only through safe accessible paths described by the manufacturer.

Key point: Scanner glass can be cleaned gently using an appropriate method.

Key point: Repeated mechanical faults should be escalated rather than forced.

Step-by-Step Guided Practical

1. Inspect paper loading on an authorised printer.
2. Open the normal printer status page.
3. Check for low-paper or consumable messages.
4. Inspect scanner glass.
5. Record one issue that requires technician support.

Ethan Digital Academy Case Study

Reliable printers and scanners support Academy registration, certificates and document workflows.

Review Questions

76. What is the purpose of printer and scanner maintenance basics?
77. State two good maintenance practices from this lesson.
78. What unsafe or unnecessary action should be avoided?

79. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 15 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 16: Maintenance Logs and Asset Records

Introduction

Maintenance records show what was checked, when it was done and what happened next.

Learning Objectives

- Explain the key principles of maintenance logs and asset records.
- Complete the safe guided practical for maintenance logs and asset records.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: A log can include date, device ID, symptom, action, result and responsible person.

Key point: Asset records can include model, serial number, RAM, storage, location and warranty.

Key point: Logs help identify repeated faults.

Key point: Do not record passwords or unnecessary personal information.

Key point: Consistent records support budgeting and replacement planning.

Maintenance Record

A simple log improves accountability

Date When work was done	Device Asset or computer name
Action What was checked or changed	Result Outcome and next action

Figure 16: Instructional visual for Maintenance Logs and Asset Records.

Step-by-Step Guided Practical

1. Create a fictional maintenance log table.
2. Add three fictional devices.
3. Record one routine update, one cleaning check and one support issue.
4. Add a status column.
5. Review the table for clarity.

Ethan Digital Academy Case Study

Ethan Digital Academy can use maintenance history to identify computers that repeatedly interrupt classes.

Review Questions

80. What is the purpose of maintenance logs and asset records?
81. State two good maintenance practices from this lesson.
82. What unsafe or unnecessary action should be avoided?
83. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 16 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 17: Safe Troubleshooting Workflow

Introduction

Troubleshooting should be systematic. The goal is to understand the problem, test safe possibilities and document the outcome.

Learning Objectives

- Explain the key principles of safe troubleshooting workflow.
- Complete the safe guided practical for safe troubleshooting workflow.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Describe the exact symptom and when it started.

Key point: Check simple external causes first: power, cables, storage, volume, display input and normal settings.

Key point: Restart when appropriate.

Key point: Use built-in diagnostic information before installing extra tools.

Key point: Change one safe factor at a time and escalate internal/electrical faults.

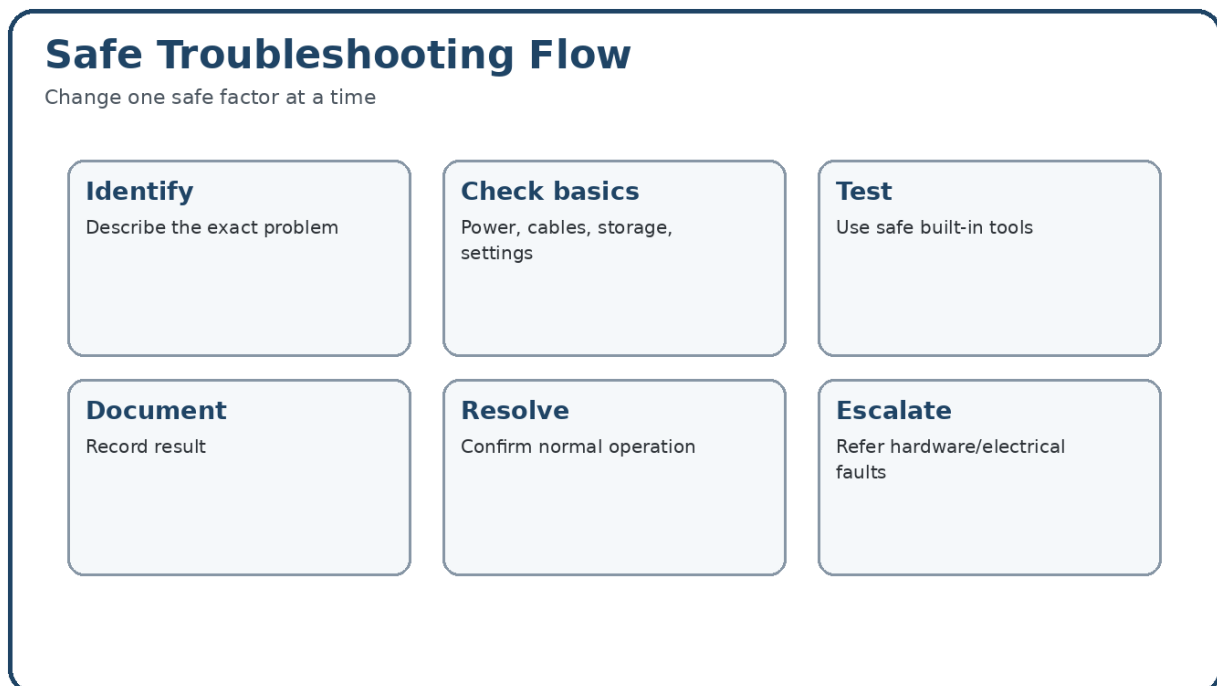


Figure 17: Instructional visual for Safe Troubleshooting Workflow.

Step-by-Step Guided Practical

1. Use a fictional 'computer is slow' scenario.
2. List observations and safe checks.
3. Use a fictional 'printer not printing' scenario.
4. List cable, queue, paper and status checks.
5. Create an escalation note for a fault you cannot safely resolve.

Ethan Digital Academy Case Study

Academy staff can reduce downtime by recording symptoms clearly before sending a device to technical support.

Review Questions

84. What is the purpose of safe troubleshooting workflow?
85. State two good maintenance practices from this lesson.
86. What unsafe or unnecessary action should be avoided?
87. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 17 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 18: Maintenance Schedule Planning

Introduction

A maintenance schedule turns occasional care into a repeatable process.

Learning Objectives

- Explain the key principles of maintenance schedule planning.
- Complete the safe guided practical for maintenance schedule planning.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: Daily checks can include obvious damage, startup and network availability.

Key point: Weekly checks may include external cleanliness, storage review and peripheral condition.

Key point: Monthly checks may include update review, backup verification and asset-log review.

Key point: Schedules should match the organisation's actual environment.

Key point: Maintenance windows should minimise disruption to teaching.

Step-by-Step Guided Practical

1. Create Daily, Weekly and Monthly columns.
2. Assign at least four tasks to each period.
3. Add a responsible role such as learner, instructor or technician.
4. Mark tasks that require administrator approval.
5. Choose a suitable maintenance window for a computer lab.

Ethan Digital Academy Case Study

An Academy lab can perform light checks between classes and deeper authorised maintenance during scheduled non-teaching periods.

Review Questions

88. What is the purpose of maintenance schedule planning?
89. State two good maintenance practices from this lesson.
90. What unsafe or unnecessary action should be avoided?
91. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 18 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Lesson 19: Final Computer Maintenance Project

Introduction

The final project asks the learner to create a complete preventive-maintenance plan for a fictional Ethan Digital Academy computer lab.

Learning Objectives

- Explain the key principles of final computer maintenance project.
- Complete the safe guided practical for final computer maintenance project.
- Know when a maintenance task should be escalated to authorised technical staff.

Teaching Content

Key point: The plan should cover hardware condition, cleaning, updates, storage, backups, security, peripherals and records.

Key point: Tasks must be assigned to suitable roles; learners should not be assigned hazardous repairs.

Key point: Include an escalation route for electrical, battery and internal hardware faults.

Key point: The plan should specify frequency and evidence of completion.

Key point: A good plan is realistic enough to follow consistently.

Final Maintenance Plan

Build a preventive-care programme

Inventory

Know the devices

Schedule

Daily, weekly and monthly checks

Backup

Protect learning and admin files

Updates

Plan approved maintenance windows

Support

Define escalation route

Records

Track actions and outcomes

Figure 19: Instructional visual for Final Computer Maintenance Project.

Step-by-Step Guided Practical

1. Create an inventory of ten fictional computers.
2. Build a daily/weekly/monthly maintenance schedule.
3. Create a one-page maintenance checklist.
4. Create a maintenance log template.
5. Write a backup and update routine.
6. Present the plan and explain which tasks require technician support.

Ethan Digital Academy Case Study

This project demonstrates how preventive maintenance supports reliable digital training operations.

Review Questions

92. What is the purpose of final computer maintenance project?
93. State two good maintenance practices from this lesson.
94. What unsafe or unnecessary action should be avoided?
95. How would this maintenance task improve reliability in a computer lab?

Assignment

Complete the Lesson 19 maintenance exercise on authorised equipment or a fictional scenario. Write 100-150 words describing the result, any risk identified and the correct next action.

Final Practical Project - 100 Marks

Component	Marks	Requirement
Lab inventory	10	Create a clear fictional device inventory.
Maintenance schedule	20	Daily, weekly and monthly tasks with suitable responsibilities.
Cleaning and inspection	10	Safe external-care checklist.
Updates and software	15	Approved update and software-hygiene routine.
Storage and backup	15	Housekeeping plus a practical backup plan.
Security maintenance	10	Basic defensive maintenance practices.
Troubleshooting/escalation	10	Safe workflow and clear stop points.
Maintenance records	10	Usable log and reporting template.

Course Quiz - 30 Questions

1. What is computer maintenance?
2. What is preventive maintenance?
3. What is corrective maintenance?
4. Why should work be saved before maintenance?
5. When should external power be disconnected?
6. Why should liquid not be sprayed into a keyboard?
7. What should be done with a damaged power cable?
8. Why are operating-system updates important?
9. Where should software updates come from?
10. Why should unknown 'booster' software be avoided?
11. Why does free storage space matter?
12. What is a backup?
13. What does the 3-2-1 backup idea encourage?
14. Why should a restore test be performed?
15. What makes a useful file name?
16. What can Task Manager or a system monitor show?
17. Why should unknown system processes not be ended randomly?
18. What is the difference between Restart and Sleep?
19. Why is sudden power loss undesirable?
20. Name two basic malware-prevention habits.
21. What can clearing browser cookies do?
22. Why should browser extensions be reviewed?
23. What should be done with a swollen laptop battery?
24. Why should laptop vents remain clear?
25. What should be checked when a printer does not print?
26. What information belongs in a maintenance log?
27. Why are asset records useful?

28. What is the first step in troubleshooting?
29. Why should only one safe variable be changed at a time?
30. Name three parts of a maintenance schedule.

Final Examination - 100 Marks

Section A - Maintenance Foundations (20 marks)

Explain preventive/corrective maintenance, safety preparation, cleaning and peripheral care.

Section B - Software and Storage Maintenance (20 marks)

Explain operating-system updates, application hygiene, storage management, file organisation and browser maintenance.

Section C - Backup and Security (20 marks)

Describe a practical backup routine, restore testing and basic malware-prevention practices.

Section D - Performance and Troubleshooting (20 marks)

Explain resource monitoring, startup/shutdown practice and a safe troubleshooting workflow.

Section E - Maintenance Planning (20 marks)

Design a maintenance schedule and record system for a fictional Ethan Digital Academy computer lab.

Course Completion Outcome

After successful completion, the learner should be able to carry out safe preventive computer maintenance, perform external cleaning and inspection, manage approved software updates, organise storage, create and test backups, monitor basic system performance, maintain browsers and peripherals, recognise issues requiring escalation, keep maintenance records and design a realistic maintenance schedule.

Professional Practice Checklist

- Save work and shut down correctly before relevant maintenance.
- Keep liquids away from equipment openings.
- Use official update and software sources.
- Back up important data before major changes.
- Do not disable security controls to force unknown software to run.
- Stop using damaged electrical equipment or swollen batteries.
- Document maintenance actions and escalate hazardous faults.